

THE HARMONIC SHEET AND THE SEXTIC ORBIT

ALEXANDER WEST TEMPLETON

ABSTRACT. Huang, Jackson, Lee, Poonen, Pries, and Zhang realize M_{23} over $\mathbb{Q}$ by a rational point in the seven-point Hurwitz fiber of covers of type $(2A, 23A, 23B)$. We determine the arithmetic of the fiber and the geometry of the rational point.

At the good prime 13 we construct an exact cover in characteristic 13 with geometric monodromy M_{23} , ordered inertia classes $(2A, 23A, 23B)$ and field of moduli $\mathbb{F}_{13^6}$, and show that the tame Hurwitz problem extends to a finite étale scheme of rank seven over $\mathbb{Z}_{13}$. Hence $H \simeq \text{Spec } \mathbb{Q} \sqcup \text{Spec } E$ with $[E : \mathbb{Q}] = 6$: the six nonrational covers form one Galois orbit, in which 13 is inert.

At the bad prime 23 we determine the stable reduction of the rational cover. It is harmonic: the tame point, the two wild points and the attachment point of the unique new tail are $0, \pm 1, \infty$ on the original component, the tails have conductors 7 and 15, and the deformation datum is $cz^6 dz/(z^{22} - 1)$ on $z^{11} = t$, the square class of c recording the classes $23A, 23B$. One nonsquare cyclotomic character reverses the line and exchanges the classes, and the harmonic datum has one lift per ordered class vector.

A descended Petri invariant names the orbit. An exact canonical model of one of the six covers over $F(\sqrt{-23})$, $F = \mathbb{Q}[x]/(x^6 - x^5 - 3x^4 - 3x^3 + x^2 + 5x + 4)$, reconstructed from the 13-adic lift of the characteristic-13 point and then verified over that field — including an exact certificate that its degree-23 map has exactly three branch points — gives a point of the Hurwitz scheme at which the invariant has minimal polynomial the recognized sextic. Hence $E \simeq F$, whose Galois closure has group S_6 and discriminant $2^4 \cdot 11 \cdot 23^4$.

The 23-adic argument identifies the local geometry of an independently known rational cover. It does not infer global rationality from local fixedness.

1. INTRODUCTION

Let H be the Hurwitz scheme of M_{23} -covers of $\mathbb{P}^1$ branched at $0, \sqrt{-23}, -\sqrt{-23}$ with inertia class $2A$ over 0 , $23A$ over $\sqrt{-23}$ and $23B$ over $-\sqrt{-23}$ (Section 2). It is a finite étale $\mathbb{Q}$ -scheme of degree seven, and [HJLPZ] exhibit a rational point $x_H \in H(\mathbb{Q})$. Two questions. Why do the other six points form one Galois orbit? What distinguishes x_H geometrically?

Two primes answer them, and the answers do not overlap. At 13, a prime of good reduction, the fiber of the tame Hurwitz scheme is one $\mathbb{F}_{13}$ -point and one closed point of degree six; this lifts, and it forces the orbit structure over $\mathbb{Q}$. At 23, the prime dividing $|M_{23}|$ exactly once, the rational cover has bad reduction of a special shape: its stable model has an original component whose four distinguished points are harmonic, and this shape is rigid enough to determine the cover among the seven once the ordered classes are fixed.

The distinction is necessary. The decomposition group at 23 is a proper subgroup of $G_{\mathbb{Q}}$, and a point of H fixed by it may lie in a $G_{\mathbb{Q}}$ -orbit of any size (Section 2.3). No 23-adic computation therefore proves that x_H is rational; rationality is the theorem of [HJLPZ]. What 23 can do is identify, among the seven, the one whose local geometry a rational point must have, and show why the symmetry that could move it does not. What 13 can do is count

orbits. Naming the six-point orbit as a number field is a third matter, settled in Part III by exhibiting one of the six covers over $F(\sqrt{-23})$ and evaluating the invariant on it.

The harmonic answer at 23 is this. After the rescaling $t = T/\sqrt{-23}$ the branch points are $0, \pm 1$, and the stable reduction of x_H has two tails, of conductors 7 and 15, attached at $t = 0$ and $t = \infty$; the deformation datum on $z^{11} = t$ is $c z^6 dz / (z^{22} - 1)$ with c determined up to squares, and the square class of c is the labelling of the wild points by 23A and 23B. The nonsquare cyclotomic characters reverse the harmonic line and exchange the two classes at once, so the ordered class vector and the datum are preserved together, and oriented patching admits exactly one lift per ordered class vector.

The answer at 13 is an exact object: a genus-4 canonical curve over $\mathbb{F}_{13^6}$ with a degree-23 map to $\mathbb{P}^1$ of passport $(23, 2^8 1^7, 23)$, whose monodromy is proved to be M_{23} by reconstructing the incidence count $23 \cdot 77 = 253 \cdot 7$ of the Witt design $S(4, 7, 23)$ from a factorization, and whose field of moduli is exactly $\mathbb{F}_{13^6}$.

The division of labor is: 23 explains the one, 13 proves 1+6, and a descended Petri function U , evaluated on an exact model of one of the six, names them.

Theorem 1.1. *Let H be as above, x_H the rational point of [HJLPPZ], and $H = \text{Spec } \mathbb{Q} \sqcup \text{Spec } E$ the decomposition induced by x_H .*

- (1) *The stable reduction of x_H at 23 is the harmonic special M_{23} -map of Theorem 9.1: original component $\mathbb{P}_t^1$ with distinguished points $0, 1, -1, \infty$, deformation datum*

$$\omega_0 = c \frac{z^6 dz}{z^{22} - 1} \quad \text{on } z^{11} = t, \quad c \in \mathbb{F}_{23}^\times / (\mathbb{F}_{23}^\times)^2,$$

primitive tail $x = u^{23} - u^9 + u^2$ of conductor 7 at $t = 0$, and new tail $y^3 = x^5 + 1$, $s = 20xy(x^5 - 8)$, of conductor 15 at $t = \infty$. The harmonic datum has exactly one lift for each ordered class vector, and x_H is the unique harmonic point of H .

- (2) *E is a field of degree six in which 13 is inert, of signature $(2, 2)$.*
 (3) *$E \simeq \mathbb{Q}[x]/(x^6 - x^5 - 3x^4 - 3x^3 + x^2 + 5x + 4)$; the Galois closure of E has group S_6 and $\text{disc } E = 2^4 \cdot 11 \cdot 23^4$.*

The theorem is not proved in its displayed order. Part I proves assertion 2 at the good prime 13; Part II proves assertion 1 at the bad prime 23; Part III proves the exact identification in assertion 3, at neither prime. In one line:

$\text{seven covers} = \underbrace{\text{one harmonic sheet}}_{23} + \underbrace{\text{one sextic orbit}}_{13, F(\sqrt{-23})}$
--

2. THE SEVEN-POINT HURWITZ FIBER

2.1. The ordered class problem. Let $G = M_{23}$, $\mathbf{C} = (2A, 23A, 23B)$, $K = \mathbb{Q}(\sqrt{-23})$, and $B = \{0, \sqrt{-23}, -\sqrt{-23}\} \subset \mathbb{P}^1$. Over K the three branch points are rational and ordered, and the Hurwitz scheme H_K of G -covers of $\mathbb{P}_K^1$ branched at B with inertia $2A$ over 0 , $23A$ over $\sqrt{-23}$ and $23B$ over $-\sqrt{-23}$ is a finite étale K -scheme with

$$H_K(\overline{K}) \simeq \text{Ni}^{\text{in}}(\mathbf{C}) = \{(x, y, z) \in 2A \times 23A \times 23B : xyz = 1, \langle x, y \rangle = M_{23}\} / M_{23},$$

a set of seven elements [HJLPPZ, Tem26].

Proposition 2.1. (1) *H_K descends to a finite étale $\mathbb{Q}$ -scheme H of degree seven.*
 (2) *H is a fine moduli scheme for G -covers in the class, and the residue field of a closed point is the field of moduli of the corresponding cover.*

- (3) For a cover Y in the class, the degree-23 quotient $X = Y/M_{22} \rightarrow \mathbb{P}^1$ has no nontrivial automorphism over $\mathbb{P}^1$, its Galois closure is Y , and it has a unique point over each of $\pm\sqrt{-23}$.

Proof. (1) Since $23 \equiv 3 \pmod{4}$, K is the quadratic subfield of $\mathbb{Q}(\zeta_{23})$, so $\sigma \in G_{\mathbb{Q}}$ swaps $\pm\sqrt{-23}$ exactly when $\chi_{23}(\sigma)$ is a nonsquare modulo 23. The same σ acts on the classes of order 23 by $y \mapsto y^{\chi_{23}(\sigma)}$, and $y^k \sim y$ in M_{23} iff k is a square modulo 23 [ATLAS], so it swaps 23A and 23B. The assignment of classes to branch points is therefore $G_{\mathbb{Q}}$ -equivariant, and it defines a descent datum for H_K over $\mathbb{Q}$. (2) $Z(M_{23}) = 1$, so a G -cover in the class has no automorphisms commuting with G ; the Hurwitz functor of G -covers with prescribed inertia is then representable [Ful69, RW06], and the residue field of a point is the field of moduli of its cover, which for $Z(G) = 1$ is also a field of definition [DD97]. (3) M_{22} is maximal and not normal in the simple group M_{23} , hence self-normalizing, so $\text{Aut}(X/\mathbb{P}^1) = N_G(M_{22})/M_{22} = 1$; it is core-free, so the Galois closure of $X \rightarrow \mathbb{P}^1$ is Y . A point over a totally ramified branch point is unique in its fiber. $\square$

2.2. The rational point and complex conjugation. [HJLPPZ] exhibit $x_H \in H(\mathbb{Q})$, the class of their printed triple; in the numbering of [Tem26] it is label 6. Complex conjugation acts on the seven as $\kappa = (1\ 2)(5\ 7)$ [Tem26, Thm. 3.1]: it fixes x_H and two other points and swaps two pairs. Write $H = \text{Spec } A$ with A an étale $\mathbb{Q}$ -algebra of degree seven. The rational point splits off an idempotent:

$$H = \text{Spec } \mathbb{Q} \sqcup \text{Spec } E, \quad \dim_{\mathbb{Q}} E = 6.$$

2.3. Why local fixedness is insufficient. The point x_H is fixed by every decomposition group, in particular by D_{23} . Nothing follows from this alone: the septic $x^7 - x - 1$ is irreducible with Galois group S_7 [Osa87], and it factors as a linear times an irreducible sextic both modulo 23 and modulo 13; so $\text{Spec } \mathbb{Q}[x]/(x^7 - x - 1)$ has a unique $\mathbb{Q}_{23}$ -point and fiber shape $1 + 6$ at 13, yet its seven geometric points form a single $G_{\mathbb{Q}}$ -orbit. In Part I the global input is the rational point x_H , which splits E off as a six-dimensional algebra; the prime 13 then shows that E is a field. In Part II the existence of a rational point is again the input, from [HJLPPZ]; the prime 23 identifies which point it is and why its local structure is rigid. Neither argument substitutes for the other.

Part 1. Six: arithmetic at 13

3. THE TAME HURWITZ MODEL

Since $-23 \equiv 4^2 \pmod{13}$, the prime 13 splits in K ; fix the place with $\sqrt{-23} \mapsto s \in \mathbb{Z}_{13}$, $s \equiv 4 \pmod{13}$. The three sections $0, s, -s$ of $\mathbb{P}_{\mathbb{Z}_{13}}^1$ are pairwise disjoint.

Lemma 3.1. *There is a finite étale $\mathbb{Z}_{13}$ -scheme $\mathcal{H}$ of rank seven with generic fiber $H \otimes \mathbb{Q}_{13}$, whose points over an algebraically closed field k of characteristic 13 are the isomorphism classes of tame M_{23} -covers of $\mathbb{P}_k^1$ branched exactly over $0, s, -s$ with inertia classes 2A, 23A, 23B respectively.*

Proof. $13 \nmid |M_{23}| = 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$, so every cover in question is tame. The Hurwitz stack of tame G -covers of $\mathbb{P}^1$ branched along disjoint sections with prescribed inertia classes is finite étale over the base [Ful69, RW06], and it is representable because $Z(G) = 1$. Its geometric fibers are the Nielsen classes, of seven elements in every characteristic prime to $|G|$: tame specialization is a bijection on inertia-marked Nielsen classes. $\square$

Frob_{13} fixes $0, \pm s$ and, since $13 \equiv 6^2 \pmod{23}$, fixes the classes $23A$ and $23B$. As $\mathcal{H}$ is finite over $\mathbb{Z}_{13}$, $\mathcal{H}(\mathbb{Q}_{13}) = \mathcal{H}(\mathbb{Z}_{13})$, and x_H reduces to a point $\bar{x}_H \in \mathcal{H}(\mathbb{F}_{13})$.

4. THE EXACT DEGREE-SIX POINT

4.1. Canonical curve and degree-23 map. Let $k = \mathbb{F}_{13}[a]/(a^6 + 12a^5 + 10a^4 + 10a^3 + a^2 + 5a + 4) = \mathbb{F}_{13^6}$; the modulus is the reduction of the sextic of Section 11. In [Rec, petri-f13] an exact smooth genus-4 canonical curve $X = V(Q, C) \subset \mathbb{P}_k^3$ is constructed, Q a quadric and C a cubic (Appendix A), together with a map

$$t = N/D: X \rightarrow \mathbb{P}^1, \quad \deg t = 23,$$

N, D quintic forms with $\text{div}(t) = 23P_0 - 23P_\infty$, $P_0 = (1:0:0:0)$. Let $\lambda \in k$ be the third branch value, put $\tau = \lambda/t$, so that τ has branch points $\infty, 1, 0$ over P_0 , the λ -fiber and P_∞ , and write $Q := P_0$.

Proposition 4.1. *The cover $\tau: X \rightarrow \mathbb{P}^1$ is tame with passport $(23, 2^8 1^7, 23)$: totally ramified over ∞ and 0, and over 1 eight places with $e = 2$ and seven with $e = 1$.*

Proof. The fiber over $t = \lambda$ is a scheme of degree 23 with reduced degree $15 = 7 + 8$ (Appendix A). The different has degree $2g - 2 + 2 \cdot 23 = 52 = 22 + 22 + 8$, so exactly the eight double points ramify, each with $e = 2$, and there is no further ramification. $\square$

4.2. Ordinary ramification points.

Proposition 4.2. *The Weierstrass gap sequence at Q and at P_∞ is $(1, 2, 3, 4)$; the first nongap is 5. In particular both totally ramified points are ordinary points of X .*

Proof. The order sequence of the canonical system at each point is $(0, 1, 2, 3)$, read off the exact local expansions of the canonical coordinates by row reduction (Appendix A); the order sequences of $|2K_X|$ and $|3K_X|$ confirm the semigroup $\{0\} \cup [5, \infty)$. $\square$

4.3. The low-pole coordinate. Since 5 is the first nongap, $L(5Q) = k \oplus k u'$. Let w be the uniformizer at Q with $\tau = w^{-23}$, unique because $\gcd(23, 13^6 - 1) = 1$, and normalize $u' = w^{-5} + O(w^{-4})$, $\text{Tr}_{k(\tau)} u' = 0$; this fixes u' .

Lemma 4.3. *u' generates $k(X)$ over $k(\tau)$. Its minimal polynomial*

$$H'(\tau, U) = U^{23} + \sum_{i \geq 5} c_i(\tau) U^{23-i} \in k[\tau][U]$$

has $\deg_\tau c_i \leq [5i/23]$, $H'(0, U) = U^{23}$, and $H'(1, U) = q'_7 d'_8{}^2$ with q'_7, d'_8 squarefree and coprime of degrees 7 and 8.

Proof. $[k(X) : k(\tau, u')]$ divides the prime 23, and $u' \notin k(\tau)$ because nonconstant elements of $k(\tau)$ have degree divisible by 23 on X while u' has degree 5. The degree bounds are the valuation bounds at Q , the constant term is $\tau = \infty$ at Q , and the shape at $\tau = 1$ is the passport. H' is computed twice, by a linear solve on the w -expansion and by Gröbner elimination (Appendix A). $\square$

5. THE WITT INCIDENCE CERTIFICATE

Let $F = k(X)$, L its Galois closure over $k(\tau)$, and

$$G_{\text{arith}} = \text{Gal}(L/k(\tau)), \quad G_{\text{geom}} = \text{Gal}(L\bar{k}/\bar{k}(\tau)),$$

the arithmetic and geometric monodromy groups, both acting on the 23 roots of H' . Since X is geometrically irreducible, G_{geom} is transitive; it is identified with $\text{Gal}(L/k'(\tau))$, $k' = L \cap \bar{k}$, a normal subgroup of G_{arith} with cyclic quotient $\text{Gal}(k'/k)$.

Over the completion at Q the roots of H' are $u'(\zeta^j w)$, $j \in \mathbb{Z}/23$, for a primitive 23rd root of unity ζ , so 7-subsets of roots are 7-subsets of $\mathbb{Z}/23$. Let $\mathcal{D}$ be the cyclic Steiner system $S(4, 7, 23)$ on $\mathbb{Z}/23$ with base block $\{0, 1, 2, 3, 9, 12, 21\}$ (Appendix B; the other cyclic $S(4, 7, 23)$ is its nonsquare twist $\mathcal{D}' = 5\mathcal{D}$), and form the block resolvent

$$\Theta(Y) = \prod_{B \in \mathcal{D}} (Y - \theta_B), \quad \theta_B = \sum_{i \in B} u'(\zeta^i w),$$

a monic polynomial of degree 253 whose coefficients are a priori Laurent series in w over $\mathbb{F}_{1366} = k(\zeta)$.

Proposition 5.1. $\Theta \in k[\tau][Y]$, with the coefficient of Y^{253-i} of degree $\leq \lfloor 5i/23 \rfloor$ in τ , and Θ is irreducible over $k(\tau)$.

Proof. Appendix B: the coefficients are computed as series to relative precision $O(w^{1450})$ and are polynomials in $\tau = w^{-23}$ within the stated bounds; irreducibility holds already over $k((\tau^{-1}))$, because the 253 roots are simple modulo w after $Y \mapsto w^{-5}Y$ and the local Galois group permutes them transitively. $\square$

Proposition 5.2. Let Ψ_{77} be the product over the 77 blocks containing 0 and Ψ_{176} the product over the other 176. Then Ψ_{77} and Ψ_{176} lie in $F[Y]$, and

$$\Theta = \Psi_{77} \Psi_{176} \quad \text{in } F[Y].$$

Proof. Appendix B: each coefficient of Ψ_{77} and Ψ_{176} is matched against the pole-order basis of $\mathcal{O}_Q = \bigcup_m L(mQ)$; the matching vanishes at the gap orders, and the difference $\Psi_{77}\Psi_{176} - \Theta$ has coefficients with poles only at Q and positive valuation there, hence zero. $\square$

Theorem 5.3. $G_{\text{arith}} = M_{23}$ and $G_{\text{geom}} = M_{23}$.

Proof. G_{arith} is transitive of prime degree, hence primitive. The inertia over $\tau = 1$ contains an element of cycle type $2^8 1^7$ (Proposition 4.1), an even permutation with seven fixed points; no affine group of degree 23 contains such an element, so by the classification of primitive groups of degree 23, $G_{\text{arith}} \in \{M_{23}, A_{23}, S_{23}\}$.

The roots of Θ are sums of roots of H' , so Θ splits in L and G_{arith} acts on its 253 roots, transitively by Proposition 5.1. By Proposition 5.2 the 77 roots of Ψ_{77} form a set stable under the stabilizer of u' , so the G_{arith} -orbit R of $\{u'\} \times \text{roots}(\Psi_{77})$ inside $\text{roots}(H') \times \text{roots}(\Theta)$ has fibers of size 77 over each root of H' :

$$|R| = 23 \cdot 77 = 1771 = 253 \cdot 7.$$

By transitivity on the roots of Θ , every fiber R_β over a root β of Θ has exactly 7 elements. The map $\beta \mapsto R_\beta$ is G_{arith} -equivariant, so its image is a single G_{arith} -orbit of 7-subsets of the roots of H' , of size at most 253. But A_{23} and S_{23} are transitive on 7-subsets, with orbit size $\binom{23}{7} = 245157$. Hence $G_{\text{arith}} = M_{23}$.

G_{geom} is a transitive normal subgroup of the simple group M_{23} , so $G_{\text{geom}} = M_{23}$. $\square$

Proposition 5.4 (orientation). *The inertia generators at Q and at P_∞ lie in different M_{23} -classes of elements of order 23. With the involution over $\tau = 1$ in the unique class $2A$, the ordered class vector of the cover is $(2A, 23A, 23B)$ up to the choice of which totally ramified point is called $23A$.*

Proof. At P_∞ , $\tau = w'^{23}$ for the exact uniformizer w' , and the roots of H' are $u'(\zeta^j w')$ with the same ζ . The block sums of the twisted design $\mathcal{D}'$, not of $\mathcal{D}$, reproduce $\Theta(w'^{23}, Y)$ (Appendix B). Transporting the P_∞ -labelling to the Q -labelling therefore composes the inertia generator with the multiplier 5, a nonsquare modulo 23, and $y^k \sim y$ in M_{23} only for square k . $\square$

The identity $23 \cdot 77 = 253 \cdot 7$ is the incidence count of points and blocks of $S(4, 7, 23)$, reconstructed from a factorization rather than assumed; A_{23} would give $22 + 231$ in place of $77 + 176$. The same certificate will settle the tails at 23 (Proposition 8.4).

6. ONE PLUS SIX

Lemma 6.1. *The cover $\tau: X \rightarrow \mathbb{P}^1$ with its inertia labelling defines a closed point of $\mathcal{H}_{\mathbb{F}_{13}}$ with residue field $\mathbb{F}_{13^6}$.*

Proof. The three branch points carry three distinct classes, so the Möbius transformation sending them, in the order $(2A, 23A, 23B)$, to $(0, s, -s)$ modulo 13 is unique, and it is defined over k because $0, \pm s$ are $\mathbb{F}_{13}$ -rational: the labelled cover is a k -point of $\mathcal{H}$. Its Frobenius conjugates are labelled covers with the same labels, and the normalized data (τ, Q, w, u') is a function of the labelled isomorphism class, so Frob^d fixes the point iff it fixes the coefficients of H' . The smallest such d is 6 (Appendix A). $\square$

Theorem 6.2.

$$\mathcal{H}_{\mathbb{F}_{13}} \simeq \text{Spec } \mathbb{F}_{13} \sqcup \text{Spec } \mathbb{F}_{13^6}$$

Consequently E is a field of degree six in which 13 is inert, with signature $(2, 2)$: $G_{\mathbb{Q}}$ acts transitively on the six nonrational points, Frob_{13} as a 6-cycle and complex conjugation as a double transposition.

Proof. $\mathcal{H}_{\mathbb{F}_{13}}$ has rank seven and contains the $\mathbb{F}_{13}$ -point $\bar{x}_H$ and the degree-six closed point of Lemma 6.1; $1 + 6 = 7$ gives the display. Idempotents of a finite étale algebra over the henselian ring $\mathbb{Z}_{13}$ lift uniquely from the special fiber, so $\mathcal{H} = \text{Spec } \mathbb{Z}_{13} \sqcup \text{Spec } \mathcal{O}$ with $\mathcal{O}$ the unramified sextic extension of $\mathbb{Z}_{13}$. Comparing with $H \otimes \mathbb{Q}_{13} = \mathbb{Q}_{13} \times (E \otimes \mathbb{Q}_{13})$, the component containing x_H is $\text{Spec } \mathbb{Z}_{13}$, so $E \otimes \mathbb{Q}_{13} = \mathcal{O} \otimes \mathbb{Q}$ is a field: E is a field and 13 is inert. The signature is read off κ , which fixes two of the six and swaps two pairs. $\square$

This proves assertion 2 of Theorem 1.1. No stable-reduction formula has been used.

Part 2. One: harmonic reduction at 23

7. THE HARMONIC COMPONENT

7.1. Collision and the exceptional line. In the coordinate T of [HJLPPZ] the three branch points are $0, \sqrt{-23}, -\sqrt{-23}$; modulo a prime above 23 they collide. The collision is a property of the coordinate, not of the configuration: after the rescaling

$$t = T/\sqrt{-23}$$

the branch points are $0, 1, -1$, and the configuration has good reduction. What is bad at 23 is the cover, whose two inertia groups of order 23 are wild. In the stable model the t -line is

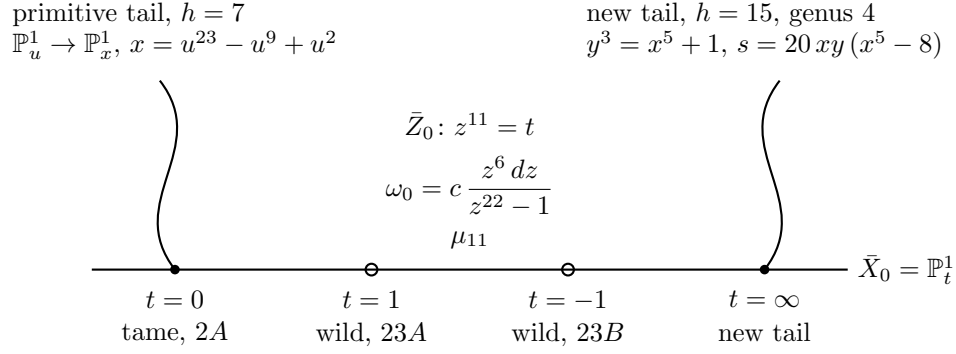


FIGURE 1. The stable reduction of x_H at 23 (Theorem 10.3). Filled points are tail attachments, open points are the wild branch points; the cross-ratio of $0, 1, -1, \infty$ is -1 . The datum lives on the μ_{11} -cover $\bar{Z}_0 \rightarrow \bar{X}_0$.

the *original component* $\bar{X}_0$ of [Wew03]; it carries the two wild points $t = \pm 1$, the primitive tail containing the tame point attaches at $t = 0$, and the unique new tail (Section 8) attaches at $t = \infty$. So $\bar{X}_0$ has four distinguished points $0, 1, -1, \infty$, of cross-ratio -1 (Figure 1).

Definition 7.1. A special M_{23} -map at 23 in the sense of [Wew03] is *harmonic* if the four distinguished points of its original component — the tame attachment point, the two wild points, and the attachment point of the new tail — form a harmonic quadruple, i.e. are $0, \pm 1, \infty$ in a suitable coordinate. A point of H is harmonic if its stable reduction at 23 is harmonic.

We use the following facts from [Wew03], for a three-point G -cover with $p \parallel |G|$ and bad reduction at $p = 23$. The stable model has an original component with a special deformation datum $(\bar{Z}_0, \omega_0)$, where $\bar{Z}_0 \rightarrow \bar{X}_0$ is the μ_m -cover with $m = |N_G(P)/C_G(P)|$, P a Sylow 23-subgroup; here $N_{M_{23}}(P) = 23:11$, so $m = 11$ and $\bar{Z}_0: z^{11} = t$. Every other component is a tail with a conductor h_j ; the primitive tail contains the tame branch point, the new tails contain no branch point, and the vanishing-cycle relation reads

$$h_{\text{prim}} + \sum_{\text{new}} (h_j - 11) = 11.$$

7.2. The primitive tail.

Proposition 7.2. *The primitive tail of x_H is $\mathbb{P}_u^1 \rightarrow \mathbb{P}_x^1$,*

$$x = u^{23} - u^9 + u^2 = u^2(u^7 + 10)^2(u^7 + 3),$$

with the tame point at $x = 0$ (fiber $2^8 1^7$), the wild point at $u = \infty$, and conductor $h_{\text{prim}} = 7$.

Proof. The exact model $F(V, T)$ of [HJLPPZ] has, at a double point of $F(V, 0)$, a Newton facet that becomes visible under the scaling $(V, T) \mapsto (23^{1/7}V, 23^{15/7}T)$; the facet polynomial is $13Z^2 + 12WZ + W^2(1 + W^7 + 2W^{14} + W^{21})$, rational with $W = w^2$, $Z = A(w^{23} + w^9) + 19w^2$, $A^2 = 7$, which is the displayed map over $\mathbb{F}_{23}$. The conductor is read off Riemann–Hurwitz: the degree-23 map $\mathbb{P}^1 \rightarrow \mathbb{P}^1$ has different of degree 44, of which 8 is tame at $x = 0$, so the wild point has different exponent $36 = (p-1)(1 + h/m)$ with $m = 11$, $h = 7$. $\square$

Proposition 7.3. *Up to isomorphism, $x = u^{23} - u^9 + u^2$ is the only tail cover with a tame point of type $2^8 1^7$ and conductor 7, and its automorphism group over the base is μ_7 .*

Proof. Appendix C: writing $x = P(u)$ with $P' = 9q$, $P = q^2r$ gives a zero-dimensional system with 429 solutions, all equivalent to the displayed one under the affine normalizations. $\square$

8. THE INFINITY TAIL

Write $G(V, S) = (1 + 23S^2)^4 F(V, 1/S)$ for the model at $T = \infty$. Its reduction at $S = 0$ is $(V + 5)^{23}$: one inseparable center. Recenter $V = 18 + W$ and let $\beta = v(S)$ vary.

Proposition 8.1 (chambers). *The Newton polygon of $G(18 + W, S)$ in W has exactly one change of lower hull as β varies, at*

$$\beta = \frac{4}{15} = \frac{23}{2 \cdot 15} - \frac{1}{2},$$

where the single wild cluster $v(W) = (3 + 6\beta)/23$ first enters $\frac{1}{5}\mathbb{Z}$. For $\beta > 4/15$ the fiber over a point of the disc splits as $3 + 5 + 15$ with $(e, f) = (3, 1), (5, 1), (15, 1)$: the tame inertia of the disc is cyclic of order exactly 15, with orbits 15, 5, 3.

Proof. The exact computation has 258 candidate breakpoints and only one change of lower hull; the ramification indices are Ore's theorem for the factors of degrees 3 and 5 and a second-order Montes step for the factor of degree 15 (Appendix C). $\square$

Corollary 8.2. x_H has exactly one new tail. It attaches at $t = \infty$, has conductor $h_{\text{new}} = 15$ and genus 4.

Proof. The tame inertia of order 15 acts faithfully on the tail through its automorphism group over the base, which is cyclic of order dividing $22h$ [Wew03, Prop. 3.11]; so $15 \mid h$. A new tail has $12 \leq h \leq 21$; hence $h = 15$, and the genus is $h - 11 = 4$. The vanishing-cycle relation $h_{\text{prim}} + (h_{\text{new}} - 11) = 7 + 4 = 11$ is exhausted, so there is no other tail. $\square$

Proposition 8.3. *The new tail is the genus-4 curve*

$$D: y^3 = x^5 + 1, \quad s = 20xy(x^5 - 8),$$

with $s: D \rightarrow \mathbb{P}^1$ of degree 23 branched only at the wild point Q over $s = \infty$, where the different exponent is $52 = (p - 1)(1 + 15/11)$.

Proof. Riemann–Hurwitz for the C_{15} -quotient with orbits of length 15, 5, 3 gives ramification $14 + 12 + 10 = 36 = 2 \cdot 4 - 2 + 30$, so the quotient has genus 0 and D is the unique cyclic 15-cover of $\mathbb{P}^1$ with these indices: $y^3 = x^5 + 1$. The degree-23 map is determined by its differential: dx/y^2 has divisor $6Q$, is exact in characteristic 23 because $dx/y^2 = y^{-23}(x^5 + 1)^7 dx$ and $5j + 1 \not\equiv 0$ for $j \leq 7$, and its primitive is unique up to constants since $l(Q) = 1$; that primitive is $s = 20xy(x^5 - 8)$, with $s^3 = 19x^3(x^5 - 8)^3(x^5 + 1)$. Its differential is dx/y^2 up to a constant, so s is unramified away from Q . $\square$

Proposition 8.4. *Both tail covers have geometric monodromy M_{23} .*

Proof. Each is transitive of prime degree; the primitive tail has inertia of type $2^8 1^7$, and the new tail is the μ_3 -pullback of $\mathbb{P}_x^1 \rightarrow \mathbb{P}_{s^3}^1$, whose inertia over 0 has type $3^6 1^5$. So each group is M_{23} , A_{23} or S_{23} . The Witt certificate of Section 5 settles both: over $\mathbb{F}_{23}(x)$ the block resolvent Θ of degree 253 is irreducible and factors over the rational function field of the tail as $\Psi_{77}\Psi_{176}$ (Appendix B), giving $23 \cdot 77 = 253 \cdot 7$ again; $D \rightarrow \mathbb{P}_s^1$ then has monodromy of index dividing 3 in M_{23} , hence M_{23} . $\square$

9. THE CENTRAL DATUM

Theorem 9.1. *The special deformation datum of x_H at 23 lives on $\bar{Z}_0: z^{11} = t$ and equals*

$$\omega_0 = c \frac{z^6 dz}{z^{22} - 1}, \quad c \in \mathbb{F}_{23}^\times,$$

where c is determined exactly up to squares.

Proof. Divisor. The datum is a logarithmic differential on $\bar{Z}_0$ with simple poles exactly at the 22 points over the wild points $t = \pm 1$, and zeros of orders determined by the conductors at the two tail attachments: $h_{\text{prim}} - 1 = 6$ at $z = 0$ and $h_{\text{new}} - 1 = 14$ at $z = \infty$. So

$$\text{div } \omega_0 = 6[0] + 14[\infty] - \sum_{\eta^{22}=1} [\eta],$$

of degree -2 on $\mathbb{P}_z^1$; the space of such differentials is one-dimensional, spanned by $z^6 dz / (z^{22} - 1)$.

Cartier. Logarithmic means $\mathcal{C}(\omega_0) = \omega_0$ for the Cartier operator. Expand at $z = 0$: $\omega_c = -c \sum_{n \geq 0} z^{6+22n} dz$. Since $\mathcal{C}(z^e dz) = z^{(e+1)/23-1} dz$ when $23 \mid e+1$ and 0 otherwise, only the terms with $23 \mid 7+22n$, i.e. $n \equiv 7 \pmod{23}$, survive; for $n = 7 + 23\ell$ the exponent $e = 6 + 22n$ has $(e+1)/23 - 1 = 6 + 22\ell$, so

$$\mathcal{C}(\omega_c) = -c^{1/23} \sum_{\ell \geq 0} z^{6+22\ell} dz = \omega_{c^{1/23}}.$$

Thus ω_c is logarithmic iff $c^{23} = c$, i.e. $c \in \mathbb{F}_{23}$. The datum is defined only up to the automorphisms $z \mapsto \xi z$, $\xi^{11} = 1$, of $\bar{Z}_0$ over $\bar{X}_0$, which send ω_c to $\omega_{c\xi^7}$; and $\{\xi^7 : \xi^{11} = 1\} = \mu_{11} = (\mathbb{F}_{23}^\times)^2$. So c is determined exactly up to its square class. $\square$

Proposition 9.2 (residues). *For $\eta^{22} = 1$, $\text{Res}_{z=\eta} \omega_c = -c\eta^7$. Over $t = 1$ the residues fill one square class of $\mathbb{F}_{23}^\times$, namely $-c \cdot (\mathbb{F}_{23}^\times)^2$; over $t = -1$ they fill the other.*

Proof. $\text{Res}_{z=\eta} = c\eta^6 / (22\eta^{21}) = -c\eta^{-15} = -c\eta^7$. Over $t = 1$ we have $\eta \in \mu_{11}$, and $\eta \mapsto \eta^7$ is a bijection of $\mu_{11} = (\mathbb{F}_{23}^\times)^2$. Over $t = -1$, $\eta = -\xi$ with $\xi \in \mu_{11}$, and $(-\xi)^7 = -\xi^7$ runs through the nonsquares. $\square$

The residue of the datum at a wild point is the character of the tame part of inertia there, and the two classes 23A, 23B are distinguished by exactly this square class ($y^k \sim y$ iff k is a square modulo 23). The square class of $-c$ is the labelling of the two wild points by the two classes.

10. ORIENTATION AND PATCHING

10.1. One lift per ordered class vector.

Proposition 10.1. *The harmonic datum with tails of conductors 7 and 15 has exactly two lifts to characteristic zero, one for each ordered class vector (2A, 23A, 23B), (2A, 23B, 23A).*

Proof. Wewers' count [Wew03, Cor. 4.8] for lifts of a special G -map is $\frac{p-1}{n'} \prod_j \frac{h_j}{d_j}$ with $n' = m = 11$ and $d_j = |\text{Aut}_G(\bar{f}_j)|$. The tail covers are unique for their conductors and have $d_j = h_j$ (Propositions 7.3, 8.3; Appendix C), so the count is

$$\frac{22}{11} \cdot \frac{7}{7} \cdot \frac{15}{15} = 2.$$

The factor $2 = [\mathbb{F}_{23}^\times : (\mathbb{F}_{23}^\times)^2]$ is the square class of c , i.e. by Proposition 9.2 the choice of which wild point is called 23A. The gluing is rigid: the conductors $7 \neq 15$ forbid exchanging the tails, the node relation forces inverse tame characters on the two sides of each node, and the tail automorphism groups μ_7, μ_{15} absorb the coordinate choices. $\square$

10.2. One nonsquare.

Proposition 10.2. *Let $\sigma \in G_{\mathbb{Q}}$ with $\chi_{23}(\sigma)$ a nonsquare modulo 23. Then σ sends $\sqrt{-23} \mapsto -\sqrt{-23}$, hence acts by $t \mapsto -t$ on the harmonic line; it exchanges $(2A, 23A, 23B)$ with $(2A, 23B, 23A)$; and on the datum it acts by $z \mapsto -z$, sending ω_c to ω_{-c} . Since -1 is a nonsquare modulo 23, the three effects are one: the pair (datum up to squares, ordered class vector) is preserved.*

Proof. The first two statements are Proposition 2.1. The involution $t \mapsto -t$ of $\bar{X}_0$ lifts to $z \mapsto -z$ on $z^{11} = t$, as $(-z)^{11} = -t$, and $\omega_c(-z) = c(-z)^6 d(-z)/((-z)^{22} - 1) = \omega_{-c}$. By Proposition 9.2 the square class of c is the labelling of the wild points, so flipping it is the same as exchanging the labels. $\square$

10.3. Harmonic comparison.

Theorem 10.3. *The stable reduction of x_H at 23 is the harmonic special M_{23} -map with datum $(\bar{Z}_0, \omega_0)$ of Theorem 9.1 and tails as in Propositions 7.2 and 8.3, and x_H is the unique harmonic point of H .*

Proof. Propositions 7.2 and 8.3 are computed on the model of x_H itself; they place the tame point at $t = 0$, the wild points at $t = \pm 1$, and the unique new tail at $t = \infty$, so the reduction is harmonic, and Theorem 9.1 gives its datum. By Proposition 10.1 the harmonic datum has one lift per ordered class vector, and the ordered class vector of x_H is fixed by the descent datum of H ; so x_H is the only point of H with this reduction. $\square$

This proves assertion 1 of Theorem 1.1. It identifies the independently known rational cover; it does not deduce rationality from 23-adic fixedness. What the argument shows is why the local structure singles out one point: the element of $G_{\mathbb{Q}}$ that could move the harmonic point reverses the harmonic line and exchanges the two wild classes at once, and oriented uniqueness makes these two operations one descent symmetry. That symmetry is the local shadow of the rationality proved in [HJLPPZ].

Part 3. The sextic coordinate

11. THE DESCENDED PETRI FUNCTION

By Theorem 6.2, E is a field of degree six. The problem is no longer connectedness; it is to name E .

11.1. Definition. Let x be a geometric point of H , X the genus-4 curve of the degree-23 cover (Proposition 2.1(3)), and $P \in X$ the point over $\sqrt{-23}$. X is not hyperelliptic (a gate of [Tem26], and automatic for a canonical complete intersection), so it is the complete intersection of its Petri quadric Q_X and a cubic. Let w be a local parameter at P with w^{23} equal to the pulled-back branch coordinate, unique up to μ_{23} , and suppose P is an ordinary point, so that $H^0(X, K_X)$ has an echelon basis $\omega_j = (w^j + O(w^4))dw$, $j = 0, \dots, 3$, with respect to w . In the dual coordinates write $Q_X = \sum_{j \leq k} q_{jk} x_j x_k$. The ratio

$$J := \frac{q_{13}}{q_{22}}$$

has weight zero: it is invariant under $w \mapsto \mu w$, hence under the μ_{23} -ambiguity and under scaling, and by [Tem26, Prop. 5.7] it is a G_K -equivariant invariant of the marked cover (X, P) , defined wherever $q_{22} \neq 0$. With τ the $K/\mathbb{Q}$ -involution acting on $A \otimes K$, the element

$$U := J + \tau J + J \cdot \tau J \in A = \mathbb{Q} \times E$$

is symmetric in the two markings, hence $G_{\mathbb{Q}}$ -equivariant: it is the coordinate $S_1 + N_1$ of [Tem26] (trace plus norm of J along $K/\mathbb{Q}$). Its characteristic polynomial on the six-dimensional $\mathbb{Q}$ -space E ,

$$g_U(x) := \det(x - m_U | E) \in \mathbb{Q}[x],$$

has degree six, and $E = \mathbb{Q}(U|_E)$ as soon as g_U is irreducible. Naming E means computing g_U .

11.2. Recognition. At x_H , $U = -5541/10237$. On the six other points, integer-relation recognition at 100 digits, replicated at a second descended coordinate with seventy digits of margin [Tem26], proposes

$$\Phi(x) = x^6 + \sum_{i=0}^5 \frac{n_i}{50019668646075143} x^i,$$

$$\begin{array}{ll} n_0 = -139453053861313073 & n_3 = -55027953306246948 \\ n_1 = -868575878997768238 & n_4 = 353749018385880145 \\ n_2 = -759255603013669319 & n_5 = 206305139624557234 \end{array}$$

and

$$\text{polredabs}(\Phi) = f := x^6 - x^5 - 3x^4 - 3x^3 + x^2 + 5x + 4.$$

Put $F := \mathbb{Q}[x]/(f)$. Φ is irreducible, $\mathbb{Q}[x]/(\Phi) \simeq F$ by the explicit embedding r of Appendix D, and F has the invariants of Section 13. Everything in this paragraph is exact arithmetic of F and Φ ; none of it says that $g_U = \Phi$.

12. EXACT REDUCTION AND RELQ

12.1. Exact evaluation at 13. The point of Section 4 is the degree-six point of $\mathcal{H}_{\mathbb{F}_{13}}$ (Theorem 6.2), and both of its totally ramified points are ordinary (Proposition 4.2), so the construction of Section 11 applies to it verbatim over k : adapted parameters w with $w^{23} =$ the branch coordinate, echelon bases, the Petri quadric, the ratio J at P_0 and at P_∞ , and $\bar{U} = J_0 + J_\infty + J_0 J_\infty$.

Proposition 12.1 (binding at 13). *On the exact cover of Section 4,*

$$\bar{U} = \bar{\theta} = 3 + 4a^2 + 2a^3 + 9a^4 + 11a^5 \in k = \mathbb{F}_{13}[a]/(f).$$

Its Frobenius orbit has length six and its minimal polynomial over $\mathbb{F}_{13}$ is

$$T^6 + 7T^5 + T^4 + 6T^3 + 7T^2 + T + 1 = \Phi \bmod 13,$$

and the embedding $r: \mathbb{Q}[x]/(\Phi) \rightarrow F$ reduces modulo 13 to $r(a) = \bar{\theta}$.

Proof. Exact computation on the model X [Rec, petri-f13]; the reduction of r is in Appendix D. $\square$

Two consequences and one boundary. Since $\Phi \bmod 13$ is irreducible, Φ is irreducible over $\mathbb{Q}$, so the recognized polynomial is primitive; and the recognized polynomial and the exact characteristic-13 geometry agree, with the right generator, at the one prime where both are available. Proposition 12.1 is a statement about the exact reduced cover. It does not by itself compare U across characteristics: it is reduction agreement at a single prime, and reduction agreement at a single prime is evidence of provenance, not proof of it. The comparison is made in Section 12.2, and it is made by exhibiting the object, not by accumulating primes.

12.2. Horizontal binding. Write $F = \mathbb{Q}[x]/(f)$ and $L = F(\sqrt{-23})$, a field of degree 12. Since $f \bmod 13$ is irreducible and $-23 \equiv 4^2$, the prime 13 has a place $\mathfrak{p}$ of L with $\beta \mapsto a$ and $\sqrt{-23} \mapsto 4$, residue field $k = \mathbb{F}_{13^6}$; write $O = O_{L,\mathfrak{p}}$.

Theorem 12.2 (RELQ). $g_U = \Phi$.

The proof exhibits a point of $H(L)$ and evaluates U at it. The data are a quadric $\hat{Q}$, a cubic $\hat{C}$ in $\mathbb{P}_L^3$ and a point $P_- \in \mathbb{P}^3(L)$; they are in the ancillary file `model-hat.gp` [Rec], and every assertion labelled (Vn) below is an exact computation with them (Appendix D). Set

$$X = V(\hat{Q}, \hat{C}) \subset \mathbb{P}_L^3, \quad P_+ = (1 : 0 : 0 : 0).$$

Proof. Step 1: the model and its reduction. (V1): the coefficients of $\hat{Q}$ and $\hat{C}$ and the coordinates of P_- are $\mathfrak{p}$ -integral, $P_\pm \in X$, and the reduction modulo $\mathfrak{p}$ is the canonical model of Section 4 written in the echelon coordinates at P_0 , with $\bar{P}_+ = P_0$ and $\bar{P}_- = P_\infty$. Let $\mathcal{X} \subset \mathbb{P}_O^3$ be the closed subscheme cut out by the integral model. It is a complete intersection in a regular scheme, hence Cohen–Macaulay, hence unmixed; every component has dimension 2, while the special fibre — the smooth genus-4 curve of Section 4 — has dimension 1. So no associated point of $\mathcal{X}$ lies in the special fibre and $\mathcal{X} \rightarrow \text{Spec } O$ is flat. Its special fibre being smooth, the non-smooth locus of $\mathcal{X} \rightarrow \text{Spec } O$ is a closed subset of $\mathcal{X}$, proper over O , whose image misses the closed point of $\text{Spec } O$; a closed subset of the spectrum of a discrete valuation ring that misses the closed point is empty. Hence $\mathcal{X} \rightarrow \text{Spec } O$ is smooth and proper and X is a smooth canonical genus-4 curve with Petri quadric $\hat{Q}$.

Step 2: a degree-23 function. (V2): the 23-jet map at $P_\pm$ has rank $21 = h^0(4K)$ on the space of quartics, so $h^0(4K - 23P_\pm) = 0$; a base point R of $|5K - 23P_+|$ would force $5K - 23P_+ \sim K + R$, hence a section of $4K - 23P_+$, so both systems are base-point-free. (V3): $\dim H^0(5K - 23P_\pm) = 4$, and the space of linear maps $M : H^0(5K - 23P_+) \rightarrow H^0(5K - 23P_-)$ with

$$M(D)D' = M(D')D \quad \text{in } H^0(10K - 23P_+ - 23P_-)$$

for all D, D' is one-dimensional. (Both sides lie in that space, and a nonzero section of $10K - 23P_+ - 23P_-$ vanishes to order at most $60 - 23 = 37$ at P_+ , so the identity is decided by the Taylor coefficients of orders 23 to 38 there.) Fix a nonzero such M . It is injective, and $M(D)D' = M(D')D$ says that $\tau := M(D)/D$ does not depend on D ; so $\tau H^0(5K - 23P_+) \subseteq H^0(5K - 23P_-)$, that is, $\text{div } \tau \geq 23P_- - 23P_+ - E_D$ for every D , where $E_D = \text{div } D - 23P_+$. Base-point-freeness makes $\bigcap_D E_D$ empty, so $\text{div } \tau \geq 23P_- - 23P_+$; both sides have degree 0, hence

$$\text{div } \tau = 23P_- - 23P_+, \quad \deg \tau = 23.$$

(V4) records the local orders of the chosen D and $M(D)$: $(23, 0)$ at P_+ and $(0, 23)$ at P_- .

Step 3: the ramification. Scale D and $M(D)$ so that their coefficient vectors on the 27 standard quintic monomials are $\mathfrak{p}$ -integral and primitive. The 27 monomials are independent modulo the ideal of the special fibre, so $\bar{D}$ and $\bar{M}(\bar{D})$ are nonzero there, $\text{div}_{\mathcal{X}} \tau$ has no

vertical component, and it is therefore the closure $23\mathcal{P}_- - 23\mathcal{P}_+$ of $\text{div } \tau$. The two sections are disjoint, since $\bar{P}_+ \neq \bar{P}_-$. At a point of $\mathcal{X}$ off $\mathcal{P}_+$ the function τ has no polar divisor through that point, so by normality it is regular there; off $\mathcal{P}_-$ the same applies to $1/\tau$. Every point misses one of the two sections, so τ is a morphism $\mathcal{X} \rightarrow \mathbb{P}_O^1$; it is proper with finite fibres, hence finite, and finite between regular schemes of the same dimension with zero-dimensional fibres, hence flat. Its fibres over 0 and ∞ are $23\mathcal{P}_\mp$, so $\deg \bar{\tau} = 23$ and, the special fibre being reduced and irreducible, $\text{div } \bar{\tau} = 23\bar{P}_- - 23\bar{P}_+$. Two functions with the same divisor on a proper curve differ by a constant, so $\bar{\tau}$ and the map t of Section 4, whose divisor is $23\bar{P}_+ - 23\bar{P}_-$, satisfy $\bar{\tau}t = c$: they differ by an automorphism of the target, and define the same cover.

τ is generically étale, so its different is an effective Cartier divisor R on $\mathcal{X}$ with $\mathcal{O}_{\mathcal{X}}(R) = \omega$, and its formation commutes with base change [Sta, Tags 0BWA, 0BW7]. Every ramification index is 23 or 2, prime to 13, so the cover is tame and R has multiplicity $e - 1$ at each ramification point [Ser79, Ch. III, §6]. Thus $\deg R = 2g - 2 + 2 \deg \tau = 52$ on both fibres, R is finite flat of degree 52 over O , and $\bar{R} = 22\bar{P}_+ + 22\bar{P}_- + \bar{J}$ with $\bar{J}$ reduced of degree 8 over the third branch value (Proposition 4.1). On the generic fibre $\text{ord}_{P_\pm} R = 22$, so $R = 22\mathcal{P}_+ + 22\mathcal{P}_- + \mathcal{J}$ with $\mathcal{J}$ finite flat of degree 8 over O and reduced special fibre: $\mathcal{J}$ is étale over O , and the generic fibre carries exactly eight further ramification points, each simple.

Step 4: three branch points. On the special fibre the eight points of $\bar{\mathcal{J}}$ lie over one value $\bar{b}$; on the generic fibre the eight points of $\mathcal{J}$ could a priori lie over several values of τ that merely coincide modulo $\mathfrak{p}$ (Remark 12.5). This step certifies that they do not. Let $z = x_1/x_0$, a local parameter at P_+ by (V5), and normalize $\tau_1 = c'\tau$ so that $1/\tau_1 = z^{23}(1 + O(z^3))$ (V6). By (V5) the point P_+ is ordinary, so $h^0(5P_+) = 2$ and there is a unique $U_0 \in L(X)$ with polar divisor $5P_+$, $U_0 = z^{-5}(1 + O(z))$ and $U_0(P_-) = 0$. (V9) computes it as $U_0 = A/x_3^2$: the osculating plane $x_3 = 0$ at P_+ cuts $3P_+ + E_3$, and A is a quadric through P_+ vanishing doubly along E_3 , the space of such quadrics being spanned by $\hat{Q}$, x_3^2 and one more form.

Both τ_1 and U_0 have poles only at P_+ , of orders 23 and 5, so $\tau_1^j U_0^k$ has pole order $23j + 5k$ there, and for $0 \leq j \leq 5$, $0 \leq k \leq 23$ these orders are pairwise distinct except for U_0^{23} and τ_1^5 . Since 23 and 5 are prime and no power of τ_1 has pole order 5 nor any power of U_0 pole order 23, neither function lies in the field generated by the other, so $[L(\tau_1, U_0) : L(\tau_1)] = 23$ and $[L(\tau_1, U_0) : L(U_0)] = 5$: the minimal polynomial of U_0 over $L(\tau_1)$ is monic of degree 23 in U and of degree 5 in τ . Its monomials of maximal pole order must cancel, so that order is 115 and its support lies in $\{23j + 5k \leq 115\}$; and at each order $m = 23j + 5k$ below 115 the coefficient h_{jk} is read off the expansion at P_+ once the coefficients of higher pole order are known. This is the ansatz; (V9) performs the back-substitution on the expansion to order z^{160} and then proves what it found. It determines

$$H_0(\tau, U) = U^{23} + \sum_{23j+5k \leq 115} h_{jk} \tau^j U^k,$$

the coefficient of z^{-m} vanishes at each of the 44 orders $m \in [0, 115]$ not of the form $23j + 5k$, and the remainder $H_0(\tau_1, U_0)$ vanishes to order z^{22} . A function with poles at most at P_+ whose expansion there has no polar part and no constant term is a constant, namely 0: so $H_0(\tau_1, U_0) = 0$ in $L(X)$. Moreover $h_{0k} = 0$ for $k < 23$ (the fibre of τ_1 over 0 is $23\mathcal{P}_-$ and $U_0(P_-) = 0$; the computation at P_+ returns this unprompted), $h_{10} \neq 0$ and $h_{50} = -1$. Thus H_0 is Eisenstein in U at the prime τ of $L[\tau]$, hence irreducible, hence the minimal polynomial of U_0 over $L(\tau_1)$; so $[L(\tau_1, U_0) : L(\tau_1)] = 23 = [L(X) : L(\tau_1)]$, $L(X) = L(\tau_1, U_0)$, and X is

the normalization of the plane curve $H_0 = 0$ with τ_1 its first coordinate. The 49 nonzero coefficients have numerators and denominators of at most 1588 digits.

(V10): there is $b \in L$, a unit at $\mathfrak{p}$, with

$$H_0(b, U) = q_7(U) d_8(U)^2 \quad \text{in } L[U],$$

where d_8 and q_7 are squarefree of degrees 8 and 7, $\gcd(q_7, d_8) = 1$ and $\gcd(d_8, \partial_\tau H_0(b, U)) = 1$; and the scalar c' of (V6) is a unit at $\mathfrak{p}$. (The witness was found by reducing modulo $\mathfrak{p}$, where the discriminant of H_0 with respect to U has degree $110 = 22 + 8 + 2 \cdot 40$ and a unique root of multiplicity 8, lifting (b, q_7, d_8) 13-adically by Newton's method, and recognizing the sixteen elements of L by lattice reduction; the displayed identities are then checked exactly, and that check is the proof.) At a root u of d_8 the point (b, u) of the plane curve is smooth with $\partial_\tau H_0 \neq 0$, so $U - u$ is a local parameter of X at the unique point above it, and $\tau_1 - b$ vanishes there to order exactly 2: these are eight points of X over b with ramification index 2. With the contributions 22 at $P_\pm$ (Step 2), Riemann–Hurwitz for the genus-4 curve X reads

$$52 = 2 \cdot 4 - 2 + 2 \cdot 23 \geq 22 + 22 + 8 = 52,$$

with equality: there is no further ramification. So the seven points over the roots of q_7 are unramified, the fibre $\tau_1^{-1}(b)$ consists of 15 points, τ is branched exactly over $\{0, b', \infty\}$ with $b' = b/c'$, and $\mathcal{J} = \tau_1^{-1}(b) \cap \{d_8 = 0\}$. Being the unique third branch point, b' is L -rational; b and c' being units at $\mathfrak{p}$, so is b' , and the branch divisor $B = \{0, b', \infty\} \subset \mathbb{P}_O^1$ consists of three disjoint sections, with $\bar{b}' = \bar{b}$.

Step 5: monodromy. Let $V = \mathbb{P}_O^1 \setminus B$. The map $\tau: \mathcal{X} \rightarrow \mathbb{P}_O^1$ is finite flat, étale over V by Steps 3 and 4, and tamely ramified along B , the indices 23 and 2 being prime to 13. By Grothendieck's specialization theory for the tame fundamental group [SGA1, Exp. XIII, 2.8–2.11], applied over the strict henselization O^{sh} : restriction gives an isomorphism $\pi_1^t(V_{\bar{s}}) \simeq \pi_1^t(V_{O^{\text{sh}}})$ and a surjection $\pi_1^t(V_{\bar{\eta}}) \rightarrow \pi_1^t(V_{O^{\text{sh}}})$, which compose to the specialization homomorphism $\pi_1^t(V_{\bar{\eta}}) \rightarrow \pi_1^t(V_{\bar{s}})$. The cover τ over $V_{O^{\text{sh}}}$ is one finite étale cover, tame along B , whose two geometric fibres are its restrictions; so the monodromy of the generic geometric fibre is the composite of the specialization map with the monodromy of the special geometric fibre. Hence the geometric monodromy groups coincide,

$$G_\eta = G_s = M_{23}$$

by Theorem 5.3, with class 2A over b' and the two classes of order 23 over ∞ and 0. The branch divisor being étale over O , the inertia classes are locally constant [Ful69, RW06] and the labelling is that of the special fibre (Proposition 5.4).

Step 6: the Hurwitz point. The Möbius transformation over L carrying $(b', \infty, 0)$ to $(0, \sqrt{-23}, -\sqrt{-23})$ — composed, if the labels come out the other way, with $z \mapsto -z$, which is the canonical isomorphism between the two class orderings — turns the cover into an M_{23} -cover of $\mathbb{P}_L^1$ branched at $0, \pm\sqrt{-23}$ with ordered class vector $(2A, 23A, 23B)$. This is a point $y \in H(L)$.

Step 7: evaluation. J depends on the adapted parameter w only through its class modulo scaling and modulo $O(w^4)$. Scaling $w \mapsto \lambda w$ scales ω_j and hence x_j by $\lambda^{-(j+1)}$ and q_{jk} by λ^{j+k+2} , and q_{13}/q_{22} is unchanged. A change $w' = w + O(w^4)$ fixes $\omega_1, \omega_2, \omega_3$ and replaces ω_0 by $\omega_0 + c\omega_3$, that is x_0 by $x_0 + cx_3$; substituting $x_0 \mapsto x_0 - cx_3$ in Q_X leaves q_{22} alone and replaces q_{13} by $q_{13} - cq_{01} = q_{13}$, so again q_{13}/q_{22} is unchanged. Both the choice of local coordinate at the branch point and the Möbius normalization of Step 6 move w within this class.

(V5): with $z = x_1/x_0$, the model satisfies $x_2/x_0 = z^2 + O(z^4)$, $x_3/x_0 = z^3 + O(z^4)$ and $\omega_0 = c(1 + O(z^3))dz$, so the coordinates of the model are the echelon basis for the parameter $z + \lambda z^4$ for a suitable λ . (V6): $1/\tau = c'z^{23}(1 + O(z^3))$; hence an adapted parameter at P_+ is $w = (c'\tau)^{-1/23} = z + O(z^4)$, and by the invariance just proved

$$J_+ = \hat{q}_{13}/\hat{q}_{22}.$$

(V7): at P_- the adapted parameter is $w' = \tau^{1/23}$; expanding the four differentials in w' , echelonizing and transforming $\hat{Q}$ accordingly gives J_- , which is the $\sqrt{-23}$ -conjugate of J_+ . Both echelon bases exist, which is to say that P_+ and P_- are ordinary points of X , as Proposition 4.2 says of their reductions. Since $U = J + \tau J + J \cdot \tau J$ is symmetric in the two markings, it does not matter which of $P_{\pm}$ is called the point over $\sqrt{-23}$. (V8):

$$U = J_+ + J_- + J_+ J_- \in F \subset L, \quad \min_{\mathbb{Q}}(U) = \Phi,$$

and in fact $U = r(\beta)$ for the embedding r of Appendix D.

Step 8: conclusion. $U(y) = U \notin \mathbb{Q}$, so y does not lie on the component $\text{Spec } \mathbb{Q}$ of H : it is a $\mathbb{Q}$ -algebra homomorphism $\iota: E \rightarrow L$, and $U(y) = \iota(U|_E)$. Then $\mathbb{Q}(U(y)) \subseteq \iota(E)$, and both have degree six over $\mathbb{Q}$, so they are equal and $U|_E$ generates E . Its characteristic polynomial is therefore its minimal polynomial: $g_U = \min_{\mathbb{Q}}(U(y)) = \Phi$. $\square$

Corollary 12.3. $E \simeq \mathbb{Q}[x]/(x^6 - x^5 - 3x^4 - 3x^3 + x^2 + 5x + 4)$, and E has the invariants of Section 13: Galois closure of group S_6 , discriminant $2^4 \cdot 11 \cdot 23^4$, sign field $\mathbb{Q}(\sqrt{11})$, and $E \otimes \mathbb{Q}_{23} = \mathbb{Q}_{23}(\sqrt{23}) \times E_4$ with E_4 totally ramified quartic.

Proof. $g_U = \Phi$ gives $E = \mathbb{Q}(U|_E) \simeq \mathbb{Q}[x]/(\Phi) \simeq F$. $\square$

Remark 12.4 (how the model was found, and why that is not the proof). The characteristic-13 point was lifted 13-adically to precision 13^{400} — the lift is unique, by a Jacobian that is invertible modulo 13 (Appendix A) — and the coefficients of $\hat{Q}$, of $\hat{C}$ and of P_- were then recognized in L by lattice reduction, with margins between 13^{75} and 13^{185} against the expected shortest vector $13^{6 \cdot 400/13}$. The quintics D and $M(D)$ of Step 2 were never recognizable: their coefficients have 7823 and 13742 decimal digits. They are not recognized; they are computed exactly over L from $\hat{Q}$ and $\hat{C}$, as kernels of jet matrices and as the solution of the symmetry system of (V3). How the model was found plays no role in the proof. Each of (V1)–(V10) is an exact computation with the printed data, and each of Steps 1–8 is a statement about the object those data define.

Remark 12.5 (the branch count is not a specialization statement). Step 4 cannot be replaced by specialization. The special fibre has three branch points, but the eight simple ramification points of the generic fibre, which specialize to the eight points over $\bar{b}$, could a priori lie over several values of τ that coincide modulo $\mathfrak{p}$; Grothendieck's theorem applies only once the branch divisor is known to be étale over O , which is what Step 4 proves. The polynomial $t = x^4 - 2x^2 + sx$ over $k[[s]]$ shows the danger: at $s = 0$ two of its three critical points share a critical value and the monodromy is D_4 , of order 8, while for $s \neq 0$ the critical values separate and the monodromy is S_4 . A draft of this proof dated 3 September 2026 argued $|G_{\eta}| \leq |G_s|$ from the Stein factorization of the Galois closure; that inequality compares G_{η} with the stabilizer of a *connected* component of the special fibre of the normalized Galois closure — a group generated by the monodromy of the special fibre together with inertia groups at points over the branch locus, since the irreducible components meet there — and not with the monodromy of the special fibre alone; it is false in the quartic example. The

gap was found by a structural review of that draft, and (V9)–(V10) were computed to close it.

Theorem 12.2 uses the prime 13 twice, and both times as geometry rather than as arithmetic: to know that the special fibre is smooth (Step 1), and to know its monodromy and its inertia labels (Steps 3–5). It uses no height bound and no second prime. The fingerprints of Theorem 6.2 — degree 6, signature (2, 2), 13 inert — are consequences now, not corroboration.

13. ARITHMETIC OF THE SEXTIC

$$f = x^6 - x^5 - 3x^4 - 3x^3 + x^2 + 5x + 4$$

degree, signature	6, (2, 2)
discriminant of f and of F	$2^4 \cdot 11 \cdot 23^4$ (index 1)
Galois group of the normal closure	S_6
sign field of $\rho_6: G_{\mathbb{Q}} \rightarrow S_6$	$\mathbb{Q}(\sqrt{11})$
2	$(e, f) = (1, 1), (1, 1), (2, 2)$
11	$(2, 1), (1, 4)$
13	$(1, 6)$: inert
23	$(2, 1), (4, 1)$: $\mathbb{Q}_{23}(\sqrt{23}) \times E_4$, E_4 totally ramified

$\text{predabs}(f) = f$, $f \equiv (x+8)^4(x+18)^2 \pmod{23}$, and the Frobenius cycle types of f and of Φ agree at each of the first 199 admissible primes (Appendix D).

Observation 13.1. $\sqrt{11}$ appears twice: as the sign field of ρ_6 and as the index $[N_{M_{23}}(C_{23}) : C_{M_{23}}(C_{23})] = 11$. We record the coincidence and do not explain it.

14. WHAT THE TWO PRIMES SAY

At 13 the fiber is 1 + 6: one $\mathbb{F}_{13}$ -point and one closed point of degree 6, and this is a theorem about H over $\mathbb{Q}$ (Theorem 6.2).

At 23 the fiber is 1 + 2 + 4 locally. The harmonic datum lifts to the singleton x_H (Theorem 10.3). The other special deformation data in this class vector are computed in Appendix C: a datum with $h_{\text{prim}} = 7$ and the new tail attached at $t \equiv \pm i$ (two lifts, over $\mathbb{Q}_{23}(\sqrt{23})$), and a datum with $h_{\text{prim}} = 8$, $h_{\text{new}} = 14$ (four lifts, conjugate over $\mathbb{Q}_{23}^{\text{ur}}(23^{1/4})$); $1 + 2 + 4 = 7$. The sextic F has $F \otimes \mathbb{Q}_{23} = \mathbb{Q}_{23}(\sqrt{23}) \times E_4$ with E_4 totally ramified of degree 4, which is what the two nonharmonic data predict for E , which by Theorem 12.2 is F . The lifts of the two nonharmonic data are not certified as the six; this paragraph is consistency, not proof.

The two primes do different work. Thirteen proves that the six are one arithmetic orbit; it says nothing about which point is the rational one. Twenty-three explains the geometry of the rational point; it cannot, by itself, force a rational point to exist. Neither argument substitutes for the other:

$\text{seven covers} = \underbrace{\text{one harmonic sheet}}_{23} + \underbrace{\text{one sextic orbit}}_{13, F(\sqrt{-23})}$
--

And the naming of the six is a third kind of work again. It happens at no prime: the object is a curve over $F(\sqrt{-23})$, and 13 enters Theorem 12.2 only to certify that its reduction is smooth and to supply the monodromy of that reduction. Witt incidence at 13, harmonic

degeneration at 23, and the exact model over $F(\sqrt{-23})$ are three separate ideas; the paper keeps them separate.

Remark 14.1 (the family $(C, 23A, 23B)$). The twisted family of [HJLPZ] makes sense for any rational class C of M_{23} in place of $2A$: class C over 0 and the two classes of order 23 over $\pm\sqrt{-23}$. Counting Nielsen classes, and testing each class fixed by complex conjugation for field of moduli $\mathbb{Q}$ through its marked Petri invariants, gives for $C = 2A, 3A, 4A, 5A, 6A, 8A$ the counts 7, 134, 660, 1422, 1632, 2272 of classes, 3, 16, 42, 32, 72, 0 of real classes, and 1, 1, 0, 0, 0, 0 of $\mathbb{Q}$ -rational ones. The $3A$ class yields a second cover over $\mathbb{Q}(T)$ with the branching of an M_{23} -cover, of genus 6, with exact models over $\mathbb{Q}(\sqrt{-23})$ and over $\mathbb{Q}$; its monodromy certificate is still owed, and the three zeros are a numerical screen with positive and negative controls. These results will appear separately with their own receipts. They are recorded here because they sharpen the question of Section 1: an explanation of the rational point of [HJLPZ] should also account for the column 1, 1, 0, 0, 0, 0.

APPENDIX A. THE EXACT CHARACTERISTIC-13 MODEL

Body statements: Propositions 4.1, 4.2, Lemmas 4.3, 6.1. This appendix records the exact objects and how each was obtained twice; the 49 coefficients of H' are printed in the ancillary file `harmonic-sheet-ancillary.pdf` and stored in `hprime-canon.txt` (SHA-256 prefix 14bc240d).

Equations. $k = \mathbb{F}_{13}[a]/(a^6 + 12a^5 + 10a^4 + 10a^3 + a^2 + 5a + 4)$. The canonical curve $X = V(Q, C) \subset \mathbb{P}_k^3$:

$$\begin{aligned} Q &= (3a^5 + a^4 + 2a^3 + 4a^2 + 4a + 2)x_0x_2 + 12x_0x_3 + (10a^5 + 12a^4 + 11a^3 + 9a^2 + 9a + 11)x_1^2 + x_1x_2 \\ &\quad + (11a^5 + 6a^4 + 7a^3 + 2a^2 + 5a + 10)x_1x_3 + x_2^2 + (3a^5 + 8a^4 + 3a^3 + 4a^2 + 2a + 7)x_2x_3 \\ &\quad + (7a^5 + 3a^4 + 3a^3 + 8a^2 + 11a)x_3^2, \\ C &= (4a^5 + 10a^4 + 2a^3 + 5a^2 + 8a + 7)x_0^2x_2 + (9a^5 + 3a^4 + 11a^3 + 8a^2 + 5a + 6)x_0x_1^2 \\ &\quad + (6a^5 + 2a^4 + a^3 + 10a^2 + 5a + 3)x_0x_1x_2 + (3a^4 + 12a^3 + 6a^2 + 4a + 2)x_0x_2^2 \\ &\quad + (7a^5 + 11a^4 + 12a^3 + 3a^2 + 8a + 10)x_1^3 + x_1^2x_2 + (11a^5 + 2a^4 + 9a^3 + 10a^2 + 9a + 4)x_1^2x_3 \\ &\quad + (10a^4 + 12a^3 + 6a^2 + 6a + 7)x_1x_2^2 + (12a^5 + 3a^4 + 11a^3 + 8a^2 + 12a + 8)x_1x_2x_3 \\ &\quad + (5a^5 + 3a^3 + 3a^2 + 7a + 8)x_1x_3^2 + (9a^5 + 8a^4 + 2a^3 + 5a^2 + 3a + 7)x_2^3 \\ &\quad + (7a^5 + 4a^4 + 11a^3 + a^2 + 10a + 4)x_2^2x_3 + (4a^5 + 8a^4 + a^3 + 7a^2 + 8a + 12)x_2x_3^2 \\ &\quad + (11a^5 + 4a^4 + 10a^3 + 6a^2 + 9a + 3)x_3^3. \end{aligned}$$

The quintics N (19 terms) and D (24 terms) with $t = N/D$ are in `exact-f13-map.json` (SHA-256 prefix 4d32f1fa); $P_\infty = (4a^5 + a^4 + 3a^3 + 12a + 7 : 6a^5 + 9a^4 + 11a^3 + 8a^2 + 12a + 2 : 9a^5 + 2a^3 + 3a^2 + 6a + 9 : 1)$ and $\lambda = 10a^5 + 9a^4 + 11a^3 + 12a^2 + 12a + 5$. The divisor $E := \text{div}(N) - 23P_0$ is reduced of degree 7 and equals $\text{div}(D) - 23P_\infty$ (Singular, saturation over $(13, a)$); the fiber over $t = \lambda$ has degree 23 and reduced degree 15.

Gaps. In the chart $x_0 = 1$ with $x_1 = z$, the coordinates $x_2(z), x_3(z)$ are power series determined by Newton iteration on $Q = C = 0$; the $4 \times N$ matrix of coordinate series has pivot columns $(0, 1, 2, 3)$, so the order sequence of $|K_X|$ at P_0 is $(0, 1, 2, 3)$ and the gaps are $(1, 2, 3, 4)$. The same holds at P_∞ in the chart $x_3 = 1$, where all four coordinates have valuation 0 and the row reduction is nontrivial. The order sequences of $|2K_X|$ (10 monomials,

dimension 9) and $|3K_X|$ (20 monomials, dimension 15) are $(0, \dots, 8)$ and $(0, \dots, 14)$ at both points, as Riemann–Roch forces from the semigroup $\{0\} \cup [5, \infty)$.

The coordinate u' . Quintics vanishing on $Z = E + 18P_0$ modulo $(Q, C)_5$ form the 2-dimensional space $\langle N, A \rangle$, and $u' = A/N$. The uniformizer w with $\tau = w^{-23}$ is $z(N/(\lambda D z^{23}))^{1/23}$ by a 23rd-root Newton iteration; u' is normalized to leading coefficient 1 against w and trace 0 (the trace is $23b_0$; the w^{23m} -coefficients of u' , $m \geq 1$, vanish as they must).

H' . $H' = U^{23} + \sum c_{i,j} \tau^j U^{23-i}$ has 49 nonzero coefficients $c_{i,j}$, $5 \leq i \leq 23$, $1 \leq j \leq \lfloor 5i/23 \rfloor$; $c_{23,5} = 12$. It was obtained twice: by a 72-unknown linear solve on the w -expansion at Q (kernel dimension 1; $H'(\tau, u')$ has poles only at Q and vanishes there to order ≥ 287 , hence is identically zero), and by Gröbner elimination of $x_0, \dots, x_3$ from $(Q, C, NU - A, N\tau - \lambda D)$ saturated by N (Singular, 5 s); all 49 coefficients agree. The smallest d with $c_{i,j}^{13^d} = c_{i,j}$ for all (i, j) is 6. $H'(1, U) = q'_7 d'_8{}^2$ with q'_7 a product of three linear and two quadratic irreducibles and d'_8 of four linear and two quadratic ones, all distinct.

APPENDIX B. THE HEPTAD CERTIFICATE

Body statements: Propositions 5.1, 5.2, 5.4, 8.4. Certificates: `mon13-heptad-certificates.json` (SHA-256 prefix 0341e61a) and `mon13-orientation.json` (83ceaf81).

The cyclic design. Relabel the stored $S(4, 7, 23)$ of [Tem26] so that the 23-cycle y_1 of the paper’s generators acts as $j \mapsto j + 1$ on $\mathbb{Z}/23$. Every block is then a 7-subset of $\mathbb{Z}/23$; the 253 blocks are the translates of 11 base blocks, and the base blocks are the square-multiples of

$$B_0 = \{0, 1, 2, 3, 9, 12, 21\}.$$

The multipliers preserving the design are exactly the 11 squares modulo 23; the nonsquare twist $\mathcal{D}' = 5\mathcal{D}$ is the other cyclic $S(4, 7, 23)$ on $\mathbb{Z}/23$, and $\mathcal{D} \cap \mathcal{D}' = \emptyset$. The block stabilizer $2^4:A_7$ has no element of order 11 or 23, so the group $\{x \mapsto qx + j : q \text{ square}\} \cong 23:11$ acts simply transitively on the blocks.

The local Galois group. $\zeta \in \mathbb{F}_{1366} = k(\zeta)$ is a primitive 23rd root of unity; $D_Q = \text{Gal}(\mathbb{F}_{1366}((w))/k((w^{23})))$ has order 253 and is generated by $w \mapsto \zeta w$ and Frob_k , which acts on ζ by $\zeta \mapsto \zeta^6$ ($13^6 \equiv 6$ generates the squares). On block sums, D_Q acts by $\theta_B \mapsto \theta_{6^m B+j}$.

Design selection and Θ . $S(4, 7, 23)$ is a 4-design with $\lambda_2 = 21$, $\lambda_3 = 5$, $\lambda_4 = 1$, so $e_1, \dots, e_4$ of the block sums do not depend on the design; e_5 and e_6 do, and only $\mathcal{D}$ gives τ -polynomials. With $\mathcal{D}$, the 253 leading coefficients $\sum_{i \in B} \zeta^{-5i}$ are pairwise distinct, and the product $\Theta = \Psi_{77} \Psi_{176}$ computed to $O(w^{1450})$ relative precision has all 253 coefficients in $k[\tau]$ within the bounds $\lfloor 5i/23 \rfloor$, with minimum absolute precision 146 beyond the constant term: 5597 nonzero coefficients over k .

Exactness of $\Theta = \Psi_{77} \Psi_{176}$. $\mathcal{O}_Q = \bigcup_m L(mQ) = k[g_5, \dots, g_9]$ with $g_5 = u'$ and $g_6, \dots, g_9 = A_m/N$ for quintics A_m vanishing on $E + 14P_0$ ($L(9Q)$ has dimension 6); every pole order $m \geq 5$ is realized by $g_5^{m/5}$ or $g_5^{\lfloor m/5 \rfloor - 1} g_{5+(m \bmod 5)}$. The coefficient of Y^{77-i} in Ψ_{77} lies in $L(5iQ)$ and is determined by its expansion from w^{-5i} to w^0 ; the matching vanishes at the gap orders $w^{-4}, \dots, w^{-1}$ for all $77 + 176$ coefficients, and the residuals vanish to full precision. This makes Ψ_{77}, Ψ_{176} elements of $\mathcal{O}_Q[Y]$. The difference $\Psi_{77} \Psi_{176} - \Theta$ has coefficients in $\mathcal{O}_Q$

with pole order ≤ 1265 , all vanishing through w^0 ; a function with poles only at Q and positive valuation there is zero.

Irreducibility of Θ . (i) After $\hat{Y} = w^5 Y$ the roots are simple modulo w , so by Hensel Θ has 253 distinct roots in $\mathbb{F}_{1366}((w))$ congruent to the θ_B , permuted by D_Q as the blocks are: transitively. Irreducible over $k((\tau^{-1}))$, hence over $k(\tau)$. (ii) Independently: at 40 random places $a \in k$, $\Theta(a, Y)$ factors with the degrees of the M_{23} permutation character on 253 blocks predicted by the cycle type of $H'(a, U)$, 40/40; there are places of pure degree 23 (five) and pure degree 11 (ten), so 253 divides the degree of any factor.

Orientation. At P_∞ , $\tau = w'^{23}$, $H'(0, U) = U^{23}$, and the roots are $u'(\zeta^j w')$ with leading coefficients $b'\zeta^j$, $b'^{23} = -c_{23,1}$, obtained by Newton on H' alone. With the same ζ , the block sums of $\mathcal{D}'$ reproduce $\Theta(w'^{23}, Y)$ in all 253 coefficients (absolute precision 701), and those of $\mathcal{D}$ fail already at e_5 . If π transports the P_∞ -labelling to the Q -labelling then $\pi m_5 \in \text{Aut}(\mathcal{D})$, so the inertia generator at P_∞ is M_{23} -conjugate to the 5^{-1} -th power of that at Q ; 5 is a nonsquare modulo 23.

The tails at 23. For $x = u^{23} - u^9 + u^2$ and for $P_{15} = 19x^3(x^5 - 8)^3(x^5 + 1)$ over $\mathbb{F}_{23}$, the degree-253 heptad relation Θ over $\mathbb{F}_{23}(x)$ is irreducible (specializations of pure degree 23 and 11), and $\Theta(P(u), Y)$ factors over the rational field $\mathbb{F}_{23}(u)$ as $\Psi_{77}\Psi_{176}$ (Singular bivariate factorization, product re-verified in PARI), with Ψ_{77} irreducible and Ψ_{176} without factors of degree 77 by parity specialization. Details in [Rec, heptad].

APPENDIX C. THE INFINITY CERTIFICATE

Body statements: Propositions 7.3, 8.1, 10.1; Section 14.

Chambers. With $V = 18 + W$ and $\beta = v(S)$, the Newton polygon of $G(18 + W, S)$ in W has vertices at $A_j(\beta) = \min_k(v_{23}(c_{jk}) + \beta k)$. The breakpoints of $\beta \mapsto \text{hull}$ are the kinks of the A_j and the collinearity points of triples (j_1, j_2, j_3) within a chamber: 258 of them, all in $[-3, 2]$; the unbounded rays $\beta > 2$ (vertex set $\{0, 3, 23\}$, slopes $1/3, 1/5$) and $\beta < -3$ (vertex set $\{0, 23\}$, slope 0) are checked algebraically. The vertex set changes exactly once, at $\beta = 4/15$, where $A_0 = 23/5$, $A_3 = 4$, $A_{23} = 0$ are collinear; below it the single cluster has $v(W) = (3 + 6\beta)/23$, which runs on $(-1/2, 4/15)$ from the original component to the tail.

Annulus and disc. At $T_0 = 1, 5, 23, 46, 529$ the fiber is a single irreducible factor of degree 23 with $v(\text{disc}) = 74, 74, 38, 38, 68$; prime degree with $f = 23$ would give $v(\text{disc}) = 0$, so $e = 23$, wild and transitive. At $T_0 = 1/23, 2/23, 1/23^2, 1/23^4$ the fiber factors as 3, 5, 15 with identical invariants. Ore: degree 3, slope $1/3$, residual $z + 9$, $(e, f) = (3, 1)$; degree 5, slope $1/5$, residual $z + 9$, $(5, 1)$; degree 15, slope $1/5$, residual $(z + 1)^3$, inconclusive. A second-order Montes step with $\phi = W^5 + 23$ gives $v_1(b_i) = 16/5, 11/5, 6/5, 0$, one segment of slope $16/15$; $\gcd(16, 3) = 1$ raises the value group from $\frac{3}{15}\mathbb{Z}$ to $\frac{1}{15}\mathbb{Z}$, so $(e, f) = (15, 1)$. Tame inertia is cyclic of order $\text{lcm}(3, 5, 15) = 15$ with orbits 15, 5, 3.

Uniqueness of the tails and $d_j = h_j$. The $h = 7$ tail is unique: $x = P(u)$ with $P' = 9q$, $P = q^2 r$, i.e. $2q'r + qr' = 9$; the zero-dimensional system has 429 solutions (Singular, squarefree eliminant of degree 232) and, on the slice $q_6 = 0$, exactly the two μ_7 -symmetric ones $q = u^8 + cu$, $c^2 = 15$, giving $w^{23} + w^9 + iw^2 \cong u^{23} - u^9 + u^2$ over $\mathbb{F}_{23}$. The $h = 15$ tail is unique by Riemann–Hurwitz as in Proposition 8.3. In M_{23} : $|N(P)| = 253$, $|C(P)| = 23$, $\{k : y^k \sim y\}$ is the set of squares; the Galois orbit of the datum has length $22 \cdot 7 \cdot 15 = 2310 = 22 \cdot \text{lcm}(7, 15)$, so it is one orbit, $d_j = h_j$, and $n' = 11$.

The other data. Two-tail special data with our class vector exist for $h_{\text{prim}} = 7, 8, 9, 10$ in numbers 3, 2, 1, 0 (`cartier.py`). Those with $h_{\text{prim}} = 7$ are the harmonic datum (new tail at $t = \infty$) and the datum with the new tail at $t \equiv \pm i$, whose two lifts have residue field $\mathbb{Q}_{23}(\sqrt{23})$; the datum with $h_{\text{prim}} = 8$, $h_{\text{new}} = 14$ has four lifts conjugate over $\mathbb{Q}_{23}^{\text{ur}}(23^{1/4})$; (9; 13) supports no M_{23} -map. $1 + 2 + 4 = 7$.

APPENDIX D. COMPUTATIONAL RECEIPTS

Software. PARI/GP 2.15.4, and 2.17.2 through cypari2 2.2.4 (gp 2.17.4 for the first stage) for (V9)–(V10); Singular 4.3 over $(13, a)$ with `minpoly`; python-flint 0.9.0 (`fq_default_poly`: a length-1450 truncated product costs 2 ms over $\mathbb{F}_{13^6}$ and 11 ms over $\mathbb{F}_{13^{66}}$); Magma V2.29 for the referee scripts of [Tem26]. Every artifact below is in [Rec]; the heptad certificate runs in 13 minutes on a 7 GB machine.

Artifacts.

claim	artifact	route
$\kappa = (12)(5\ 7)$	[Tem26], kappa certificate	GAP, Magma
model X , $t = N/D$, passport	<code>petri-f13/exact-f13-map.json</code>	Sage, exact
gaps (1, 2, 3, 4)	<code>mon13/canon_gaps.py</code> , <code>canon_2k.py</code>	Python
E , Z , $L(9Q)$	<code>mon13/canon_E.sing</code> , <code>canon_L9.sing</code>	Singular
H' , series	<code>mon13/m13a_canon.gp</code> $\rightarrow$ <code>hprime-canon.txt</code>	PARI
H' , elimination	<code>mon13/elim_H.sing</code>	Singular
cyclic $S(4, 7, 23)$	<code>mon13/cyclic_design.py</code>	Python
Θ , $\Psi_{77}\Psi_{176} = \Theta$	<code>mon13/heptad13.py</code> $\rightarrow$ <code>heptad-certificate.json</code>	python-flint
orientation	<code>mon13/orient13.py</code> $\rightarrow$ <code>mon13-orientation.json</code>	python-flint
datum, Cartier	<code>harmonic/</code> , <code>cartier.py</code>	PARI
chambers, Ore, Montes	<code>harmonic/newton_chambers.py</code> , <code>inf_montes.gp</code>	PARI
tails E_0 , D , s	<code>tails/</code>	Singular, PARI
M_{23} for the 23-tails	<code>heptad/</code>	Singular, PARI
Φ , r , invariants of F	[Tem26] receipts; <code>m23-sexitic/</code>	flint, PARI, Magma
Petri value mod 13	<code>petri-f13/petri-binding-mod13.json</code>	Sage, exact
13-adic lift to 13^{400}	<code>relq/lift13b.py</code> $\rightarrow$ <code>liftb-400.json</code>	python-flint
$\hat{Q}$, $\hat{C}$, P_- in W_{400}	<code>relq/canon13.py</code> $\rightarrow$ <code>canon-400.json</code>	python-flint
recognition in L	<code>relq/recon_canon.py</code> $\rightarrow$ <code>reconCanon-400.json</code>	flint lll
the model over L	<code>relq/model-hat.gp</code>	—
(V1), reduction	<code>relq/check_V1_f13.py</code> vs <code>petri-f13/</code>	PARI, exact
(V1)–(V8)	<code>relq/run_verify.gp</code> , <code>verify_L.gp</code>	PARI, exact
(V9): U_0 , H_0	<code>eightpoint/stage2.py</code> $\rightarrow$ <code>eightpoint-H0.txt</code>	PARI, exact
(V10): b , q_7 , d_8	<code>eightpoint/stage3.py</code> $\rightarrow$ <code>eightpoint-witness.txt</code>	PARI, exact

SHA-256 prefixes: Θ certificate 0341e61a, H' 14bc240d, orientation 83ceaf81, exact map 4d32f1fa, model over L 7423ca42, H_0 (eightpoint-H0.txt) df9eb6d1, eight-point witness (eightpoint-witness.txt) 8380a45e.

The exact model over L . An element of $L = F(\sqrt{-23})$ is written $(m(\beta) + \sqrt{-23}n(\beta))/d$ with $m, n \in \mathbb{Z}[x]$ of degree < 6 and $d \in \mathbb{Z}_{>0}$. In these coordinates $\hat{Q}$ has $\hat{q}_{00} = \hat{q}_{01} = 0$, $\hat{q}_{03} = -1$, $\hat{q}_{02} = -\hat{q}_{11}$, $\hat{q}_{12} = \hat{q}_{22} = 1$, and

$$J_+ = \hat{q}_{13} = \frac{m_+(\beta) + \sqrt{-23}n_+(\beta)}{4601809515438913156},$$

$$\begin{aligned}
m_+ &= 428507493019923191x^5 - 756316978528056623x^4 - 1016504823899978721x^3 \\
&\quad + 235256771917067587x^2 + 366369501472437623x + 1063669928380756003, \\
n_+ &= 219051448639587523x^5 - 662132559244474971x^4 + 266418207781929355x^3 \\
&\quad - 256772595495992321x^2 + 145546733094649219x + 462232824183065315,
\end{aligned}$$

and J_- is the $\sqrt{-23}$ -conjugate of J_+ . The remaining coefficients — $\hat{q}_{11}$, $\hat{q}_{23}$, $\hat{q}_{33}$ (up to 53 digits), the twenty coefficients of $\hat{C}$ (up to 122 digits) and the three coordinates of P_- (up to 53 digits) — are in `model-hat.gp`. The verification `run_verify.gp` runs in 150 seconds on a 7 GB machine; the longest step is the exact solution of the symmetry system (V3) over L . The three rows of the table for the 13-adic lift and the recognition document how the model was found (Remark 12.4); their scripts are in `relq/`, and their outputs are superseded by `model-hat.gp`, which is the object of the proof.

The eight-point certificate. The script `stage1-ND.gp` recomputes D , $M(D)$ and c' as in `run_verify.gp`; `stage2.py` with the library `lser.gp` produces H_0 ; `stage3.py` with `stage3-lib.gp` produces and checks the witness; `replay.sh` runs the three. The series arithmetic of (V9) is done in a representation of $L((z))$ that keeps integral coefficients with one denominator per series, over the primitive element $w = \sqrt{-23} - \beta$ of L (minimal polynomial $w^{12} + 2w^{11} + 133w^{10} + \dots + 194457104$); a product of two series of length 160 then costs a few seconds, where the nested representation of `run_verify.gp` costs minutes. The expansion of X at P_+ to order 160, the normalization of τ , the construction of U_0 and the back-substitution take 12 minutes on two cores; the lift and recognition of (V10) at precision 13^{2500} take 150 seconds, with recognition margins of at least 13^{478} against the expected $13^{6 \cdot 2500/13}$. The element b has coordinates of 395 digits over a denominator of 386 digits; the coefficients of q_7 and d_8 have up to 606 and 692 digits; h_{10} , the largest coefficient of H_0 , has 1588. Modulo $\mathfrak{p}$ the discriminant of H_0 in U factors as $b^{22}(b - \bar{b})^8 N(b)^2$ with N of degree 40 — the double roots consistent with the 40 nodes of the plane model — in Frobenius orbits of sizes 1, 3, 3, 9, 24; and $H_0(\bar{b}, U)$ factors as $q_7 d_8^2$ with d_8 of type $1^4 2^2$ and q_7 of type $1^3 2^2$ over $\mathbb{F}_{13^6}$. The exact checks of (V10) — the identity $H_0(b, U) = q_7 d_8^2$, the two squarefreeness and the two coprimality statements, and the unit tests for b and c' — take 9 seconds.

The field F . $f = x^6 - x^5 - 3x^4 - 3x^3 + x^2 + 5x + 4$ has $\text{disc } f = \text{disc } F = 2^4 \cdot 11 \cdot 23^4$ (index 1), signature (2, 2), $\text{polgalois} = S_6$, $\text{polredabs}(f) = f$, quadratic subfield of the Galois closure $\mathbb{Q}(\sqrt{11})$. Local data (e, f) : 2: (1, 1), (1, 1), (2, 2); 11: (2, 1), (1, 4); 13: (1, 6); 23: (2, 1), (4, 1), with $f \equiv (x + 8)^4(x + 18)^2 \pmod{23}$. Frobenius inventory over the first 199 primes prime to $2 \cdot 11 \cdot 23$ and to the denominator of Φ : cycle types [6]: 44, [1, 5]: 41, [2, 4]: 24, [3, 3]: 8, [1, 1, 4]: 20, [1, 2, 3]: 36, [2, 2, 2]: 3, [1, 1, 1, 3]: 12, [1, 1, 2, 2]: 11, identical for f and Φ . The embedding $\mathbb{Q}[x]/(\Phi) \rightarrow F$ is $\alpha \mapsto r(\beta)$ with

$$\begin{aligned}
50019668646075143 \cdot r &= 7568631896021296 x^5 - 29458505838561912 x^4 \\
&\quad + 22903915474596992 x^3 + 25097895504098152 x^2 \\
&\quad + 14793393724122520 x - 61970328454788839,
\end{aligned}$$

$\Phi(r(\beta)) = 0$ exactly, and $r(a) \equiv 11a^5 + 9a^4 + 2a^3 + 4a^2 + 3 = \bar{\theta} \pmod{13}$.

AI assistance and verification. This project made extensive use of AI systems, principally Claude (Anthropic) and ChatGPT (OpenAI): exploratory discussion, proposing and criticizing approaches, literature triage, writing and debugging code, running and analysing

computations, and drafting this text under the author’s direction. The author chose the questions, decided which claims to pursue, supervised the computations, evaluated the evidence, and is responsible for every claim. All computational claims are reproducible from the named artifacts; AI-generated suggestions were not treated as mathematical evidence.

REFERENCES

- [HJLPPZ] X. Huang, B. Jackson, K.-H. Lee, B. Poonen, R. Pries, S. Zhang, *The Mathieu group M_{23} is a Galois group over $\mathbb{Q}$* , preprint (2026), arXiv:2608.08538.
- [Tem26] A. W. Templeton, *The Galois action on the seven M_{23} Hurwitz points*, preprint (2026), DOI 10.5281/zenodo.22073397; artifacts DOI 10.5281/zenodo.22063041.
- [Rec] A. W. Templeton, *Artifacts for “The Harmonic Sheet and the Sextic Orbit”*, GitHub repository <https://github.com/azide0x37/m23-harmonic-sheet>, archived at Zenodo, DOI 10.5281/zenodo.XXXXXXXX (2026). Directories `relq/`, `eightpoint/`, `petri-f13/`, `mon13/`, `harmonic/`, `tails/`, `heptad/`, `m23-sextic/`.
- [Wew03] S. Wewers, *Three point covers with bad reduction*, J. Amer. Math. Soc. **16** (2003), 991–1032, arXiv:math/0205026.
- [Ful69] W. Fulton, *Hurwitz schemes and irreducibility of moduli of algebraic curves*, Ann. of Math. **90** (1969), 542–575.
- [RW06] M. Romagny, S. Wewers, *Hurwitz spaces*, in *Groupes de Galois arithmétiques et différentiels*, Sémin. Congr. **13**, Soc. Math. France (2006), 313–341.
- [DD97] P. Dèbes, J.-C. Douai, *Algebraic covers: field of moduli versus field of definition*, Ann. Sci. École Norm. Sup. (4) **30** (1997), 303–338.
- [Osa87] H. Osada, *The Galois groups of the polynomials $X^n + aX^l + b$* , J. Number Theory **25** (1987), 230–238.
- [Sta] The Stacks Project Authors, *The Stacks Project*, <https://stacks.math.columbia.edu>, 2026.
- [SGA1] A. Grothendieck, M. Raynaud, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Math. **224**, Springer, 1971; Documents Mathématiques **3**, Soc. Math. France, 2003.
- [Ser79] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics **67**, Springer, 1979.
- [ATLAS] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker, R. A. Wilson, *Atlas of Finite Groups*, Oxford Univ. Press, 1985.

INDEPENDENT RESEARCHER. ORCID 0009-0005-7058-3912

Email address: alex@templeton.cloud